MIL-I-46414A(MU)
3 MAY 1965
SUPERSEDING
MIL-I-46414 (ORD)
27 OCTOBER 1960

MILITARY SPECIFICATION

INITIATOR, BURSTER, M13, PARTS FOR AND LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the parts, loading, assembling and packing for one type of burster initiator designated as Burster Initiator, M13 for use in smoke cartridges for 4.2 inch mortars.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 — Preservation; Methods of.

MIL-A-2550 — Ammunition and Special Weapons; General Specification for.

MIL-P-10025 — Packaging, Packing and Marking for Interplant Shipment of Inert Ammunition Components, General Specification for.

MIL-I-45607 — Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).

MIL-STD-109 — Quality Assurance Terms and Definitions.

MIL-STD-1235 — Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ORDNANCE CORPS

7548324 — Carton, Packing, Ammunition, for Initiator, Burster, T7 Assembly, Details, Packing and Marking.

FSC 1315

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7548325 — Box, Packing, Ammunition, for Initiator, Burster T7 Assembly, Details and Packing.

8796522 — Marking Diagram and Sealing for Wood Packing Boxes.

8797844 — Initiator, Burster, M13 Assembly.

PUBLICATIONS

ORDNANCE CORPS

IEL–8797844 — Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Assembly. The burster initiator shall comply with all requirements specified on Drawing (dwg.) 8797844 and with all requirements specified in applicable specifications.

3.3 Workmanship. All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, cracks, sharp edges, unblended radii, dirt, grease, flux, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any parts nor shall the parts be contaminated by the cleaning agent. Surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define terms used herein. The provisions of Specification MIL–A–2550 shall apply.

4.1.1 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all of the quality assurance provisions specified within this specification.

(b) Number of units of product inspected.

(c) Results obtained, by defect code, for all inspections performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Certificates of compliance on all material purchased by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization.

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The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Initial production inspection

4.2.1 At the start of regular production, the contractor shall submit an initial production sample to a Government approved facility as designated by the contracting officer (see 6.4) for evaluation in accordance with the provisions of 4.2.2. The initial production sample shall consist of twenty-five (25) complete sets of parts (i.e., twenty-five (25) each of every component (inert components only)) which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of analysis for materials. An initial production sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer.

4.2.2 *Inspections to be performed.* Components will be subjected by the Government to any or all of the examinations or tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings.

4.2.3 *Rejection.* If any component fails to comply with any of the applicable requirements, the initial production sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new initial production sample or portion thereof. Until an initial production is accepted, the contractor is in no way authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.3 Inspection provisions

4.3.1 *Lot formation.* A lot shall consist of burster initiators produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance or interchangeability, as determined by the Government, shall not necessitate changing the lot number. Lots shall be

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submitted for inspection in accordance with Standard MIL–STD–105 (or Standard MIL–STD–1235, when applicable). Each lot shall contain.

(a) Tetryl from not more than one lot from one manufacturer.

(b) Parts of one lot interfix number from one manufacturer.

(c) Loaded detonators from one manufacturer in accordance with the same drawing, same drawing revision, same specification and same specification revision of one lot interfix number.

4.3.2 *Examination.* Inspection for Critical defects shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with MIL–STD–105 except that continuous sampling plans in accordance with Standard MIL–STD–1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect. Equipment necessary for the performance of the Inspections listed shall be in accordance with 4.3.4.

4.3.2.1 *Cup* (see dwg. 8797845 covering a detail of dwg. 879. 9797844).

Categories	*Defects*	*Method of inspection*	*Code No. (see 6.2)*
Critical: None defined			
Major: None defined			
Minor:	AQL 2.5 percent		
201.	Inside diameter	Gage	01001
202.	Outside diameter	Gage	01002
203.	Length	Gage	01003
204.	Wall thickness through base	Gage	01004
205.	Evidence of poor workmanship (see 3.3)	Visual	01005

4.3.2.2 *Cushion, Detonator* (see dwg. 8797846 covering a detail of dwg. 8797844).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major: None defined			
Minor:	AQL 1.5 percent		
201.	Inside diameter	Gage	02001
202.	Outside diameter	Gage	02002
203.	Thickness	Gage	02003
204.	Evidence of poor workmanship (see 3.3)	Visual	02004

4.3.2.3 *Disc* (see dwg. 8797847 covering a detail of dwg. 8797844).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major: None defined			
Minor:	AQL 2.50 percent		
201.	Diameter	Gage	03001
202.	Thickness	Gage	03002
203.	Evidence of poor workmanship (see 3.3)	Visual	03003

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4.3.2.4 *Ring, snap* (see dwg. 8797917 covering a detail of dwg. 8797844).

Categories	Defects	Method of inspection	Code No.
Critical: None defined			
Major: None defined			
Minor:	AQL 2.50 percent		
201.	Thickness	Gage	04001
202.	Distance between ends	Gage	04002
203.	Distance from end to outside surface	Gage	04003
204.	Diameter	Gage	04004
205.	Radius	Gage	04005
206.	Evidence of poor workmanship (see 3.3)	Visual	04006

4.3.2.5 *Body* (see dwg. 8797850 covering a detail of dwg. 8797844).

Categories	Defects	Method of inspection	Code No.
Critical: None defined			
Major:	AQL 1.00 percent		
101.	Diameter of pellet cavity	Gage	05001
102.	Diameter of detonator cavity	Gage	05002
103.	Depth to bottom of detonator cavity	Gage	05003
104.	Depth of countersink	Gage	05004
105.	Diameter of flash hole	Gage	05005
106.	Depth of pellet cavity	Gage	05006
107.	Radius missing	Visual	05007
Minor:	AQL 4.00 percent		
201.	Location and shape of Snap Ring groove	Gage	05008
202.	Location of crimping groove	Gage	05009
203.	Thickness of flange	Gage	05010
204.	Total length	Gage	05011
205.	Outside diameter of shank	Gage	05012
206.	Diameter of crimping groove	Gage	05013
207.	Distance to flat on flange rim	Gage	05014
208.	Finish improper	Visual	05015
209.	Protective finish damaged exposing base metal	Visual	05016
210.	Evidence of poor workmanship (see 3.3)	Visual	05017

4.3.2.6 *Assembly, prior to crimping detonator* (see dwg. 8797844).

Categories	Defects	Method of inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Detonator cushion missing	Visual	06001
Minor: None defined			

4.3.2.7 *Assembly, prior to crimping cup* (see dwg. 8797844).

Categories	Defects	Method of inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Pellet missing	Visual	07001
102.	Felt pad missing	Visual	07002

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Categories	Defects	Method of inspection	Code No.
Minor:	AQL 1.50 percent		
201.	Diameter of pellets, minimum (min.)	Gage	07003
202.	Length of pellets, min.	Gage	07004
203.	Evidence of poor workmanship (see 3.3)	Visual	07005

4.3.2.8 *Assembly* (see dwg. 8797844).

Categories	Defects	Method of inspection	Code No.
Critical: None defined			
Major:	AQL 1.0 percent		
101.	Detonator missing	Visual	08001
102.	Detonator inverted	Visual	08002
103.	Snap ring missing	Visual	08003
104.	Detonator loose	Manual	08004
105.	Cup crimp, not 360 degrees and not snug	Visual Manual	08005
106.	Detonator crimp, not 360 degrees and not pressed firmly against detonator	Visual Manual	08006
Minor:	AQL 2.5 percent		
201.	Diameter at cup end, maximum (max.)	Gage	08007
202.	Length, max.	Gage	08008
203.	Marking misleading or unidentifiable	Visual	08009
204.	Protective coating damaged in handling, to the extent of exposing base metal	Visual	08010
205.	Evidence of poor workmanship (see 3.3)	Visual	08011

4.3.2.9 *Carton, prior to sealing* (see dwg. 7548324).

Categories	Defects	Method of inspection	Code No.
Critical: None defined			
Major: None defined			
Minor:	AQL 1.5 percent		
201.	Any packing material missing	Visual	09001
202.	Initiator missing	Visual	09002
203.	Tray damaged to the extent that initiators touch each other	Visual	09003

4.3.2.10 *Carton* (see dwg. 7548324).

Categories	Defects	Method of inspection	Code No.
Critical: None defined			
Major: None defined			
Minor:	AQL 2.5 percent		
201.	Sealing strip torn, incomplete, badly wrinkled or otherwise fails to seal carton	Visual	10001
202.	Carton torn, cut or punctured	Visual	10002
203.	Marking missing, misleading or unidentifiable	Visual	10003
204.	Contents loose	Manual	10004

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4.3.2.11 *Sealed barrier bag* (see dwg. 7548324).

Categories	Defects	Method of inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Bag punctured, torn or cut	Visual	11001
102.	Seal mislocated or incomplete	Visual	11002
Minor: None defined			

4.3.2.12 Packing box, prior to sealing (see dwg. 7548325).

Categories	Defects	Method of inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Carton missing	Visual	12001
Minor: None defined			

4.3.2.13 *Packing box, sealed* (see dwg. 7548325).

Categories	Defects	Method of inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Box broken, board loose or protruding nail	Visual	13001
102.	Hardware or strapping missing, broken or loose	Visual	13002
Minor:	AQL 1.5 percent		
201.	Marking misleading or unidentifiable	Visual	13003
202.	Hardware or strapping improperly engaged or mislocated	Visual	13004
203.	Contents loose	Visual	13005

4.3.3 *Testing.*

4.3.3.1 Heat seal test of barrier (see dwg. 7548324) — Major defect — Code No. 11004. The sampling plan for this test shall be in accordance with Standard MIL-STD-105, Code letter H and AQL 0.25 percent. The test shall be performed as specified in 4.4.1.

4.3.4 *Inspection Equipment.* Index of Inspection Equipment Number IEL-8797844 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL I-45607.

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4.3.4.3 *Government use of contractor's inspection and test equipment.* The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures

4.4.1 *Heat seal test of barrier bag.* The test procedure for this test shall be as prescribed in Specification MIL–P–116 for the "Heat Seal Test".

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 *Level A.* The loaded initiators shall be packaged in accordance with Level A requirements of dwg. 7548324.

5.1.2 *Level C.* The loaded initiators shall be packaged in accordance with Level C requirements of dwg. 7548324.

5.2 Packing.

5.2.1 *Parts*

5.2.1.1 *Level C.* Packing and marking for parts shall be in accordance with Specification MIL–P–10025.

5.2.2 *Loaded*

5.2.2.1 *Level A.* The cartons shall be packed in accordance with the Level A requirements of dwg. 7548325.

5.2.2.2 *Level C.* The cartons shall be packed in accordance with the Level C requirements of dwg. 7548325.

5.3 Marking. Marking shall be in accordance with dwgs. 7548325, 7548325 and 8796522.

5.4 Data cards. Data cards shall be prepared for each lot in accordance with information specified in Specification MIL–A–2550.

6. NOTES

6.1 Ordering data. Procurement documents shall specify title, number and date of this specification.

6:2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 *Inspection equipment lists.* Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (00 Form 1242–3) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3,

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6.3.4, 6.3.5 and 6.3.6 will be designated in the IEL by the following codes:

CDAF — Contractor's design responsibility on Army fromat in accordance with MIL–D–45608.

CDCF — Contractor's design responsibility on contractor format.

AD — Army design.

ADMU — Army design, mandatory for use.

CE — Commercial equipment.

SCD — Specification control drawing.

6.3.3 *Army designs.* Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and non-mandatory. When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use coc-parable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Specification control drawing.* Specification control drawings depict the maximum equipment requirements in outline, descriptive, diagrammatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 *Commercial equipment.* Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor design identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDAF shall comply with the format and requirements of MIL–D–456, and in addition, contain the detail and information specified above. Final approved contractor designs identified as CDCF shall be suitable for microfilming. They must be non-reproducible diazo or electrostatic prints of top quality full size

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with black line on white background and shall provide a legible print produced via the 35MM microfilm process. Contractor shall submit one copy of the final design as a rolled set or flat set.

6.3.7 *Submission of contractor designs.* Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 Submission of initial production samples. Instructions as to the location for evaluation of initial production shall be obtained from the contracting officer. Upon receipt of such request, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly.

6.5 Supersession data. This specification supersedes in part MIL–I–10291B dated 24 November 1958, covering Initiators, Burster, M1, M2 and T7 Loading, Assembling and Packing.

Custodian:

Army—MU

Preparing activity:

Army—MU

Project No. 1315–A446

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